Conditional Probability and Bayesian Inference

From Updating Beliefs to Building Intelligent Systems

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MAI 600: Mathematics and Statistics for AI

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Lecture Outline

Conditional Probability

Bayes' Theorem

Bayesian Inference

Applications in AI

Why Study This for AI?

1. Language Models (LLMs)

How does a model like GPT decide the next word?

`"The cat sat on the ___."`

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It computes the **conditional probability** of every word in its vocabulary, given the preceding words.

- $P(\text{"mat"} | \text{"The cat sat on the"}) = 0.85$
- $P(\text{"chair"} | \text{"The cat sat on the"}) = 0.10$
- $P(\text{"rocket"} | \text{"The cat sat on the"}) = 0.00001$

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2. Spam Filtering

How does your email client know an email is spam?

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How does your email client know an email is spam?
It uses **Bayes' Theorem** to calculate the probability that an email is spam, given the words it contains.

$P(\text{Spam}|\text{contains "prize", "free"})$

The Core Idea: Updating Probabilities with New Info

Motivating Question

Imagine a standard deck of 52 cards.

- What is the probability of drawing a King?

$$P(\text{King}) = \frac{4}{52} \approx 7.7\%$$

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- **Given this new information**, what is the probability it's a King?

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$$P(\text{King} \,|\, \text{Face Card}) = \frac{4}{12} \approx 33.3\%$$

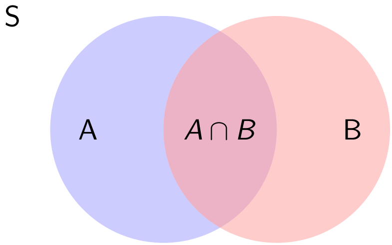
This is **conditional probability**: the probability of an event A occurring, *given* that another event B has already occurred.

Formal Definition

The conditional probability of event A given event B is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where $P(B) > 0$.



Interpretation: We are reducing our "universe" of possibilities from the entire sample space S down to just the outcomes in event B.

Example: Revisiting the Joint PMF

Building on the last lecture: Let's use our joint PMF for two fair coin flips.

	Y=0 (Tails)	Y=1 (Heads)	**P(X)**
X=0 (Tails)	0.25	0.25	**0.5**
X=1 (Heads)	0.25	0.25	**0.5**
P(Y)	**0.5**	**0.5**	1.0

Let's calculate the probability that the first coin is Heads ($X = 1$) **given** that the second coin is Heads ($Y = 1$).

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$$= \frac{0.25}{0.5} = 0.5$$

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This makes sense! Since the coin flips are independent, knowing the outcome of the second coin doesn't change the probability of the first.

The Chain Rule of Probability

By rearranging the definition of conditional probability, we get a useful tool for calculating the probability of intersections:

$$P(A \cap B) = P(A|B)P(B)$$

This can be extended to multiple events:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \ldots$$

Example: Drawing Two Cards (No Replacement)

What is the probability of drawing a King, then another King?

- Let A be "first card is a King". $P(A) = 4/52$.
- Let B be "second card is a King".

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We want to find $P(A \cap B)$. Using the chain rule:

$$\begin{aligned} P(A \cap B) &= P(A) \times P(B|A) \\ &= \frac{4}{52} \times \frac{3}{51} \approx 0.45\% \end{aligned}$$

From Conditional Probability to Bayes' Theorem

We know two things:

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Rearranging this gives us the famous **Bayes' Theorem**:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

This simple formula is the foundation for an entire field of statistics and AI, allowing us to "invert" conditional probabilities.

The Terminology of Bayes' Theorem

$$\underbrace{P(H|E)}_{\text{Posterior}} = \frac{\overbrace{P(E|H)}^{\text{Likelihood}}\overbrace{P(H)}^{\text{Prior}}}{\underbrace{P(E)}_{\text{Evidence}}}$$

- H: Hypothesis (the thing we want to know, e.g., "patient has disease")
- E: Evidence (the data we observe, e.g., "test is positive")
- **Prior** $P(H)$: Our belief in the hypothesis *before* seeing any evidence.
- **Likelihood** $P(E|H)$: The probability of seeing the evidence *if the hypothesis is true*.
- **Posterior** $P(H|E)$: Our *updated* belief in the hypothesis *after* seeing the evidence.
- **Evidence** $P(E)$: The overall probability of observing the evidence. This acts as a normalization constant.

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A rare disease affects 1 in 10,000 people. A test for this disease is 99% accurate (99% true positive rate, 99% true negative rate). A patient tests positive. What is the probability they actually have the disease?

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Let's define our events:

- H: Patient has the disease.
- E: Patient tests positive.

What are our probabilities?

- **Prior** $P(H) = 1/10000 = 0.0001$
- **Likelihood** $P(E|H) = 0.99$ (True positive)
- **False Positive Rate** $P(E|\neg H) = 1 - 0.99 = 0.01$

We need to find the **Posterior**: $P(H|E)$.

Example: Medical Diagnosis (Calculation)

First, we need the denominator, $P(E)$, using the Law of Total Probability:

$$\begin{aligned} P(E) &= P(E|H)P(H) + P(E|\neg H)P(\neg H) \\ &= (0.99)(0.0001) + (0.01)(0.9999) \\ &= 0.000099 + 0.009999 \\ &= 0.010098 \end{aligned}$$

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Conclusion: Even with a 99% accurate test, a positive result only means there's a 1% chance the patient has the disease. This is because the **prior** probability was so incredibly low. This counter-intuitive result

The Big Idea: From Numbers to Distributions

In the previous example, our belief (the prior) was a single number. But what if our belief isn't a single number, but a range of possibilities?

Bayesian Inference is the process of using Bayes' theorem to update our beliefs about a model's parameters. Our "belief" is represented by a probability distribution.

The Old Way

The coin is fair.

$p = 0.5$

The Bayesian Way

I'm not sure if the coin is fair.
My belief about its fairness is captured by this distribution.

The Bayesian Inference Process

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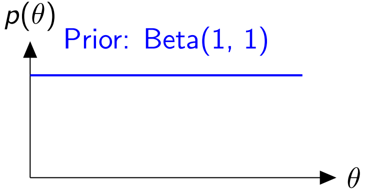
3. **Collect Data** and apply Bayes' Theorem.
4. **Calculate the Posterior Distribution**:

$$p(\theta|\text{Data}) \propto p(\text{Data}|\theta)p(\theta)$$

The posterior distribution is our *updated* belief about θ after considering the data. It is a compromise between our prior belief and the evidence from the data.

Example: Is this Coin Fair?

- **Parameter** θ: The true probability of getting Heads.
- **Prior** $p(\theta)$: We have no idea, so let's assume any value of θ between 0 and 1 is equally likely. This is a Uniform distribution, which is also a Beta(1, 1) distribution.
- **Data**: We flip the coin 10 times and get 7 Heads, 3 Tails.
- **Likelihood** $p(\textbf{Data}|\theta)$: The number of heads in n flips follows a Binomial distribution.



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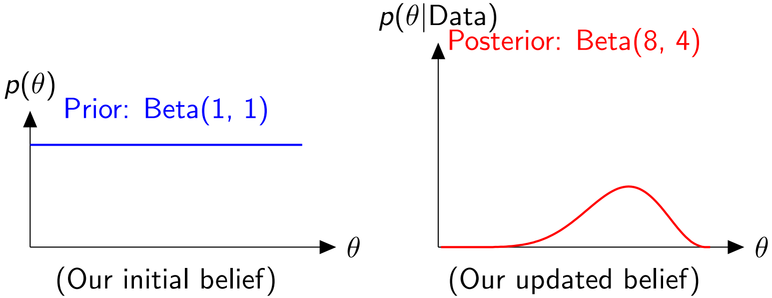

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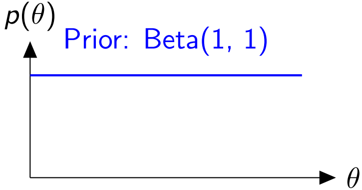

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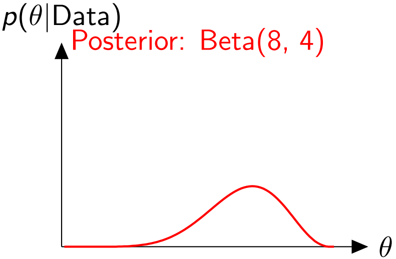

(Our updated belief)

Our new belief is now peaked around $\theta = 0.7$, but it still shows uncertainty.

AI Application 1: Bayesian A/B Testing

The Old Way (Frequentist)

Run an experiment for a fixed sample size, calculate a p-value. The result is a binary "significant or not". It's hard to interpret and you can't peek at the results.

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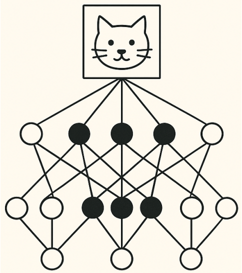
The Bayesian Way

Model the conversion rates for versions A and B as probability distributions. Update these distributions as data comes in.

- **Intuitive Results**: We can directly calculate the probability that B is better than A, $P(\theta_B > \theta_A)$.
- **Faster Decisions**: We can stop the test as soon as we are confident enough, saving time and resources.
- **Flexibility**: The posterior from one experiment can become the prior for the next one.

Standard Neural Network

Learns a single, fixed value for each weight.



Prediction: "This is a cat."

AI Application 2: Bayesian Neural Networks

Standard Neural Network

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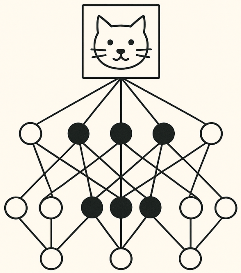
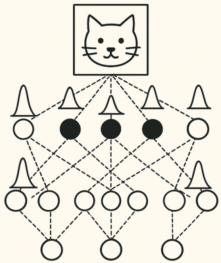


Prediction: "This is a cat."

Bayesian Neural Network (BNN)

Learns a full probability distribution for each weight.



Prediction: "This is a cat with 95% *confidence*."

AI Application 3: Probabilistic Programming

The Challenge

Calculating the posterior distribution by hand is only possible for simple models. For complex AI models, the math becomes intractable.

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The Solution: Probabilistic Programming Languages (PPLs)

Languages like **PyMC**, **Stan**, or **Pyro** let you build Bayesian models programmatically.

- You just **define the model**: specify the priors and the likelihood.
- The PPL's inference engine automatically computes the posterior distribution for you, using powerful algorithms like Markov Chain Monte Carlo (MCMC).

Lecture Summary

- **Conditional Probability**: $P(A|B) = P(A \cap B)/P(B)$, the probability of A given B has occurred. It's the engine behind sequential data models like LLMs.
- **Bayes' Theorem**: A simple way to invert conditional probabilities, allowing us to update our beliefs in a hypothesis given new evidence.
- **Bayesian Inference**: Extends Bayes' theorem to update entire probability distributions for model parameters, moving from a single "best" answer to a full representation of our uncertainty.
- **AI Applications**: From smarter A/B testing to safer neural networks that know when they're uncertain, Bayesian methods are a cornerstone of modern, robust AI.

Thank you. Questions?